AMRO AL-BAALI

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SKILLS

Domain Knowledge: Sensor fusion, state estimation, localization, SLAM, pose graph optimization, computer vision, multi-sensor extrinsic calibration, matrix Lie groups, linear algebra.
Programming Languages: C++, Python, MATLAB.
Libraries and Tools: ROS, Eigen, Numpy, OpenCV, Ceres, GTSAM, NvBlox, cuRobo (mapper).

EDUCATION

M. Eng Mechanical **McGill University**

05/2019 - 08/2021 Montreal, Canada

- CGPA: 3.77/4.00. Supervisor: Prof. J. R. Forbes.
- Thesis title: *Augmenting Sensor Measurements with INS Estimates in a Graph Based SLAM Problem*.
- Publication: A. Al-Baali, T. Hitchcox and J. R. Forbes, "Combining DVL-INS and Laser-Based Loop Closures in a Batch Estimation Framework for Underwater Positioning," in *IEEE Journal of Oceanic Engineering*, doi: `10.1109/JOE.2023.3286854`.

B. Eng Honours Mechanical, Minor in Computer Science **McGill University**

09/2014 - 04/2019 Montreal, Canada

- CGPA: 3.83/4.00. Supervisor: Prof. J. R. Forbes.
- Thesis title: *Parallel Feedforward Control Using Linear Matrix Inequalities*.

WORK EXPERIENCE

Senior SLAM Engineer & Team Lead - Scene Perception **Pickle Robot**

02/2024 – Present Cambridge, MA

- Designed and implemented the real-time obstacles (3D) mapping stack that generated ESDFs used by Motion Planning and control to generate and execute collision-free arm trajectories. The stack fused 3D lidars and package perception estimates leveraging libraries such as NvBlox, cuRobo (mapper), and KISS-ICP, generating a dense ESDF for consumption by the Motion Planning team.
- Designed and prototyped a real-time detection and tracking system that estimates the 3D pose of telescopic conveyors with respect to the AMR's conveyor, enabling the navigation to align the robot's conveyor with the telescopic conveyor.
- Led the design, development, and production deployment of a complete localization and semantic mapping system shipped to customer sites, combining 2D lidar odometry with semantic mapping of container walls and ramps, robust against wheel slip while the robot arm picks packages.
- Tech lead for the Scene Perception team (4–5 engineers), driving multi-modal sensor extrinsic calibration, scene perception, and obstacles mapping, and the sub-system architecture.
- Built a sensor data playback pipeline decoupling perception algorithm development from hardware, enabling rapid, reproducible iteration on localization and State Estimation without physical robot access.

Software Engineer - Robotics Team **Vention**

05/2023 – 01/2024 Montreal, Canada

- Designed and implemented a new feature that brought a new types of users (power users) that are interested in directly controlling the robot through programming it. I worked on the architecture that allows the user to simulate the commands before executing them on the robot, allowing for faster prototyping for the customer.
- Worked closely with customers to introduce the new feature to them and customize it for their needs. I provided periodic updates including prototypes the customers can test on.
- Improved the process of prototyping on **ROS**, which was used to run the robot simulations, by automating the build and deployment processes. My changes reduced friction in the development process, allowing developers to prototype more quickly and to introduce less bugs.

Software Engineer - Localization and Mapping **Avidbots**

09/2021 – 03/2023 Kitchener, Canada

- Contributed to the maintenance and development of lidar-based localization and mapping pipeline, improving the non-linear least squares algorithms used in state estimation and scan-matching.
- Contributed to the development of new features, including localization close to walls (enabling wall following), and visual-based localization for improved initial-localization.
- Led the effort of adapting wheel odometry pipeline to accommodate the removal of wheel encoders to reduce the robot cost. This involved developing a new calibration routine to calibrate the steering wheel encoders and changing the kinematic model.
- Led the effort of setting up an OptiTrack system to evaluate the localization performance during in-house tests. The effort included the design and implementation of a novel calibration routine that aligned the OptiTrack coordinate system with the map used in the localization pipeline.

Graduate Student - SLAM **DECAR group (McGill University)**

05/2019 – 08/2021 Montreal, Canada

- Collaborated with Voyis and Sonardyne to develop a SLAM algorithm for an Autonomous Underwater Vehicle (AUV) equipped using a third-party Inertial Navigation System (INS) and the Voyis Insight Pro high-precision laser scanner.
- Researched and designed a novel localization protocol that treats the INS as a black box and fuses its output with Voyis' high-precision laser scanner, which was used to generate highly precise maps of the seabed used by Voyis for various applicaations. The algorithm leveraged convex and on-manifold optimization.

Mechanical Engineering Intern **MY01**

05/2018 – 04/2019 Montreal, Canada

- Designed and executed mechanical tests on the MY01 device to obtain medical certification, including FDA certification.
- Wrote the software that communicates and controls the testing platform, including moving the test head for penetration testing, generating plots, and writing a GUI for the user to control the device.
- Customized the Autodesk Valut software tool used for CAD version control, customizing it to automatically generate reports and maintain version numbers for released designs.

Undergraduate Research Assistant **McGill University**

05/2017 – 08/2017 Montreal, Canada

- Researched the use of multi-objective optimization methods, such as genetic algorithms, for for design-space exploration (DSE), under the supervision of Prof. D. Varro.
- Used OpenMDAO and Platypus packages to implement the optimization for the DSE.

Teaching Assistant **McGill University**

09/2017 – 04/2021 Montreal, Canada

I was a teaching assistant for the following courses, providing tutorials, grading assignments, mid-terms and quizzes, and assisting to students to understand the course materials and assignments.

- MECH 513 (Control Systems) (Winter 2021)
- MECH 309 (Numerical Methods) (Fall 2019)
- MECH 412 (System Dynamics and Control) (Fall 2017)